AKSHAYKUMAR KATHIRVELU

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PROFESSIONAL SUMMARY

Security professional with 3+ years of experience across network security, security operations, and information security. Investigated 90+ incidents monthly using Splunk and Wireshark packet analysis, reducing MTTR by 80%. Diagnosed critical enterprise outages, including one affecting 10,000+ users, resolved 100+ vulnerabilities, and developed cloud and endpoint detections to identify security monitoring gaps.

TECHNICAL SKILLS

SIEM & Security Operations: Splunk (SPL), Microsoft Sentinel (KQL), Microsoft Defender for Endpoint, Sigma, MITRE ATT&CK
Cloud & IaC: AWS (IAM, CloudTrail, CloudWatch), Azure (Entra ID, Defender for Cloud), GCP (Security Command Center), Terraform
Network Security: Firewall and ACL policies, 802.1X, RADIUS, Wireshark, TCP/IP, DHCP, OSPF, BGP, SSL/TLS
Vulnerability, Risk & Compliance: Nessus, Qualys, Nmap, NIST, CIS Benchmarks
OS & Scripting: Windows, Linux, Python, SQL, PowerShell, Bash, Git

CERTIFICATIONS

CompTIA Security+ | ISC2 Certified in Cybersecurity | AWS Certified Cloud Practitioner | AWS Certified AI Practitioner | Microsoft Certified: Azure Fundamentals | Fortinet NSE 1, NSE 2 & NSE 3

PROFESSIONAL EXPERIENCE

SECURITY ANALYST | Comtech LLC **May 2025 – Aug 2025**

- Transformed MSP capabilities into a standardized MSSP offering, defining alert triage, endpoint monitoring, gap assessment, escalation workflows, and response playbooks across 5+ client environments.
- Identified 23+ inactive accounts and 15+ excessive admin/write privileges across 500+ devices through NIST-aligned gap assessments and SIEM/Defender investigations, documenting remediation to reduce access-control exposure.
- Developed Comtech's CMMC compliance framework, mapping control requirements to service capabilities; applied STRIDE/DREAD threat modeling and authored 3 technical whitepapers used in a client proposal.

NETWORK SECURITY ENGINEER (Tier 1 to Tier 2) | Movate (HPE-Aruba Networking) **Sep 2021 – Jul 2024**

- Investigated 90+ network and security incidents monthly, including priority escalations across 5,000+ network devices supporting 3,000+ customers, using root-cause analysis (RCA) and Nmap diagnostics to isolate routing, DHCP, and authentication failures.
- Diagnosed and corrected a BGP ASN mismatch to restore customer WAN connectivity and isolated an HA node synchronization failure that caused an OSPF routing-table mismatch and RADIUS authentication failure, restoring network access for 10,000+ users.
- Reduced MTTR by 80%, from hours to minutes, by correlating centralized syslog and security-event data in Splunk against Wireshark packet captures, isolating root cause without reproducing the issue.
- Resolved 100+ vulnerabilities identified through 200+ Qualys and Nessus assessments, prioritizing findings by severity and exploitability.
- Implemented and troubleshot 802.1X, RADIUS, IPsec VPNs, and firewall/ACL policies, resolving connectivity failures while mentoring 20+ Tier 1 engineers on incident investigation and network hardening.
- Automated recurring network operations using Python, PowerShell, and Bash for backups, configuration changes, and log collection.

TEACHING EXPERIENCE

GRADUATE TEACHING ASSISTANT (Information Security) | George Mason University **Aug 2025 – May 2026**

- Troubleshot Kali/Metasploit security labs for 200+ Information Security students across 4 sections, analyzing packet captures, Windows Event Logs, and IOCs to resolve payload and reverse-shell failures.
- Authored lab guides and troubleshooting walkthroughs for 6 security labs per semester, documenting repeatable investigation steps and helping students resolve common issues independently.

PROJECTS

Microsoft Defender Threat Hunting & Detection, KQL | Defender for Endpoint | GitHub **Sep 2026**

- Executed a 6-step living-off-the-land chain across 8 MITRE ATT&CK techniques on a monitored Windows 11 endpoint without alerts; ran 6 KQL hunts across 4 telemetry tables to uncover the activity and convert validated logic into a scheduled custom detection.

AWS CloudTrail Threat Detection, CloudWatch | Sigma | MITRE ATT&CK | GitHub **Aug 2026**

- Built and validated 3 CloudTrail detections using CloudWatch metric filters and alarms for root usage, IAM privilege escalation, and log tampering; measured 84–208-second alert delivery and identified a gap where 11 matching events produced only 4 notifications.

EDUCATION

Master of Science in Applied Information Technology (Cybersecurity) | George Mason University **Aug 2024 – May 2026**
Bachelor of Engineering in Electronics and Communication | Anna University **Aug 2017 – May 2021**